1. What is Time Value of Money (TVM)?

Simple Idea:

> **"Money today is more valuable than the same money tomorrow."**

The **Time Value of Money** is the idea that **money today is worth more than the same amount in the future**, due to its potential earning capacity. This fundamental principle of finance recognizes that:

> "A rupee today is worth more than a rupee tomorrow."

Why?

- You can **use** today's money to **invest**, **buy**, or **earn more**.
- **Prices increase** over time (due to inflation), so ₹100 today might not buy the same things in the future.
- Future is uncertain — who knows what might happen tomorrow?

Real-Life Example:

Imagine your friend offers you ₹500 now or ₹500 after one year.
You would take it **now**, right?
Why? You could:

- Spend it now
- Save it and earn interest
- Or just not take the risk of losing it later!

2. What is Future Value (FV)?

Simple Idea:

> **Future Value** tells you how much your money will grow if you save or invest it.
>
> Future Value is the amount to which an investment grows over time when it is invested at a certain interest rate. **Compounding** means **earning "interest on interest"**.

Example:

You put ₹1,000 in a savings account. Bank gives 10% interest per year.

After 1 year: ₹1,000 + 10% = ₹1,100
After 2 years: ₹1,100 + 10% = ₹1,210
(This is called **compounding** — earning on your interest!)

Formula:

For one-time investment:

$$\mathrm{FV} = PV \times (1 + r)^n$$

Where:

- PV = Present Value (initial investment)
- r = interest rate per period
- n = number of periods

Example:

If you invest ₹1,000 at 10% for 2 years:

FV=1000×(1+0.10)2=1000×1.21=₹1210

3. What is Present Value (PV)? What is Discounting?

Simple Idea:

> **Present Value** tells you what a future amount is worth **today**.
>
> Present Value is the current worth of a future sum of money, given a specified rate of return. **Discounting** is the reverse of compounding.
>
> **Going backward** from future money to find out what it's worth **today**.

Think of it as the **reverse of compounding**.

Formula:

$$\mathrm{PV} = \frac{FV}{(1+r)^n}$$

Example:

If you will receive ₹1,000 after 1 year, how much is that worth today?
Maybe ₹900 — because you could invest ₹900 today and turn it into ₹1,000 next year.

So instead of waiting for ₹1,000 next year, you would be happy with ₹900 now.

What is the present value of ₹1,000 received 3 years later if the discount rate is 8%?

$$PV = \frac{1000}{(1+0.08)^3} = \frac{1000}{1.2597} = ₹793.83$$

4. What is Compounding?

Simple Idea:

> **Earning interest on your interest**.

Example:

You invest ₹1,000. You earn 10% interest yearly.

- Year 1: ₹1,000 → ₹1,100
- Year 2: ₹1,100 → ₹1,210
- Year 3: ₹1,210 → ₹1,331

Every year, your money grows faster because you are earning on your previous earnings too.

6. What is an Annuity?

Simple Idea:

> A **fixed amount of money** you receive (or pay) **every month or year**.

An **annuity** is a series of **equal payments** made at regular intervals over time.

There are two types:

- **Ordinary Annuity**: Payments at the **end** of each period.
- **Annuity Due**: Payments at the **beginning** of each period.

Formula:

Future Value of an Ordinary Annuity:

$$FV = P \times \frac{(1+r)^n - 1}{r}$$

Present Value of an Ordinary Annuity:

$$PV = P \times \frac{1 - (1+r)^{-n}}{r}$$

Where:

- P = Periodic Payment
- r = Interest rate per period
- n = Number of periods

Example:

You win a lottery that gives you ₹10,000 **every year** for 5 years.

That's an **annuity**.

Or when you take a **loan** and repay **every month**, that's also an annuity.

Investing ₹5,000 every year at 10% for 3 years:

$$FV = 5000 \times \frac{(1+0.10)^3 - 1}{0.10} = 5000 \times 3.31 = ₹16,550$$

7. What is a Perpetuity?

Simple Idea:

A payment that **never ends**. It goes on **forever**!

Formula :

$$PV = \frac{P}{r}$$

Where:

- P = Annual payment
- r = Interest rate

Example:

You buy a special bond that pays you ₹1,000 every year **forever**.

You're getting money endlessly — this is a **perpetuity**.

A security pays ₹1,000 annually forever, and the interest rate is 5%:

$$PV = \frac{1000}{0.05} = ₹20,000$$

Why it matters in Corporate Finance:

Concept	Why Companies Use It
Time Value of Money	To compare money today vs future for decisions

Future Value	To know how much investments grow
Present Value	To know what future earnings are worth today
Annuities	To calculate loan repayments or investment returns
Perpetuities	To value stocks or fixed income forever

Final Thought:

Understanding time value of money helps you make **smarter financial decisions**, whether it's saving, investing, or borrowing — both in **personal life** and **business**.

Concept	Formula	Key Use
Future Value	$FV = PV(1+r)^n$	To find future worth of present money
Present Value	$PV = \frac{FV}{(1+r)^n}$	To assess value of future cash today
Annuity (PV)	$PV = P \times \frac{1-(1+r)^{-n}}{r}$	Loans, EMIs, investments
Perpetuity	$PV = \frac{P}{r}$	Valuing infinite cash flows

1. Effective Annual Rate (EAR)

What is it?

- The **Effective Annual Rate** (EAR) is the **real interest rate earned (or paid)** on an investment or loan **after compounding** is taken into account.

- It reflects the **true cost of borrowing** or the **real return on investment**.

Why does it matter?

- Interest is often **compounded more than once a year** (monthly, quarterly, etc.).
- So, 10% annual interest compounded monthly is actually more than 10% over the year — EAR tells us **how much more**.

🔹 **Formula:**

$$EAR = \left(1 + \frac{i}{n}\right)^n - 1$$

Where:

- i = nominal interest rate (APR)
- n = number of compounding periods per year

🔹 **Example:**

If APR is 12%, compounded monthly:

$$EAR = \left(1 + \frac{0.12}{12}\right)^{12} - 1 = 12.68\%$$

2. Annual Percentage Rate (APR)

🔹 What is APR?

The **Annual Percentage Rate (APR)** is the **yearly interest rate charged on a loan** or earned on an investment, **excluding the effect of compounding**.
It is also called the **nominal rate**.

Key Points:

- It represents **only the interest cost**, **not the compounding frequency**.
- It is often the **rate advertised by banks** or lenders.
- APR is used for comparison **only when compounding is not frequent or consistent**.

🔹 Why is APR used?

- APR gives a **standard way to quote interest rates** so consumers and businesses can compare financial products.
- Regulatory bodies (e.g., RBI, SEC) often require lenders to disclose APR for **transparency**.

Formula:

There is **no compounding** included in APR:

APR=Periodic Interest Rate×Number of Periods per Year\text{APR} = \text{Periodic Interest Rate} \times \text{Number of Periods per Year}APR=Periodic Interest Rate×Number of Periods per Year

Example:

- Monthly rate = 1%
- APR = 1%×12=12%1\% \times 12 = 12\%1%×12=12
-

But this doesn't account for compounding — if you invest ₹100 at 1% monthly, you'll earn **more than ₹12 in a year**.

Example-
Let's say you are offered a **loan of ₹1,00,000 at 12% APR**, compounded monthly.

Step 1: APR is 12%

Monthly rate = 12% ÷ 12 = 1% = 0.01

Step 2: Compute EAR

$$EAR = \left(1 + \frac{APR}{n}\right)^n - 1$$

Where:

- $APR = 0.12$
- $n = 12$

$$EAR = \left(1 + \frac{0.12}{12}\right)^{12} - 1 = (1.01)^{12} - 1 = 1.1268 - 1 = 0.1268 = 12.68\%$$

✅ So, **APR = 12%**
✅ But **EAR = 12.68%**

That means you are actually paying **12.68% effective interest**, not just 12%.

Corporate Finance Implications of APR

1. **Loan Cost Comparison**
    - A company comparing two loans: one with **12% APR monthly compounding**, and one with **11.75% APR quarterly compounding**.
    - EAR will show **which is actually cheaper** after considering compounding.
2. **Lease or Borrow Decisions**
    - APR helps compare the **stated rates** offered by equipment financiers.
    - EAR helps determine the **real financial burden**.
3. **Capital Budgeting**
    - When discounting cash flows, using APR could understate opportunity cost.
    - EAR is more appropriate for **NPV / IRR decisions**.
4. **Investor Perspective**
    - APR on bonds or fixed deposits is the **headline rate**.
    - Investors should calculate EAR to know **real return**.

Summary Table

Term	Meaning	Includes Compounding?	Use
APR	Annual Percentage Rate	❌ No	For quoting loans/investments
EAR	Effective Annual Rate	✅ Yes	For comparing real returns or costs

3. Applications of APR and EAR in Investment Decisions

APR (Annual Percentage Rate) and EAR (Effective Annual Rate) are **critical tools** in evaluating and comparing the **true cost or return** on financial products. Here's how they're used in real-world decision-making:

1. Comparing Loans

Context:

Companies often need to take loans (for capital expenditure, expansion, or working capital). Multiple banks may offer loans with different **interest rates and compounding frequencies**.

Problem:

Bank A offers 11.5% APR compounded **monthly**
Bank B offers 11.8% APR compounded **quarterly**
Which loan is actually cheaper?

Solution:

Calculate **EAR** for both to compare true cost:

- **Bank A:**

$$EAR = \left(1 + \frac{0.115}{12}\right)^{12} - 1 = 12.15\%$$

- **Bank B:**

$$EAR = \left(1 + \frac{0.118}{4}\right)^{4} - 1 = 12.33\%$$

Conclusion: Even though Bank B's APR looks slightly higher, the compounding effect makes it **more expensive** in reality. So, Bank A is cheaper when considering EAR.

2. Evaluating Investments

Context:

A company has the option to invest surplus cash in one of the following:

- Investment A: 10% interest **annually**
- Investment B: 10% interest **semi-annually**

Problem:

Both investments seem to offer the same return — but compounding plays a key role.

Solution:

- Investment A EAR = 10% (no compounding)
    - Investment B:

$$EAR = \left(1 + \frac{0.10}{2}\right)^{2} - 1 = 10.25\%$$

Conclusion: Investment B is **more profitable** due to **semi-annual compounding**.

This is essential for:

- Treasury teams deciding where to park funds.
- Comparing bonds or fixed deposits with varying payment intervals.

3. Discounting Future Cash Flows (Capital Budgeting)

Context:

When a company evaluates a new project (e.g., setting up a factory), it forecasts **future cash flows**. To determine if the project is worth it, it uses **Discounted Cash Flow (DCF)** techniques like:

- **Net Present Value (NPV)**
- **Internal Rate of Return (IRR)**

Problem:

If the wrong discount rate (APR instead of EAR) is used, the NPV may be **under or overestimated**.

Solution:

Use **EAR as the discount rate** when cash flows are frequent (monthly, quarterly).

🧮 Example:
Project cash inflows are received **quarterly**, and the APR is 12%.
To properly discount, convert to **EAR**:

$$EAR = \left(1 + \frac{0.12}{4}\right)^4 - 1 = 12.55\%$$

This gives a **more accurate present value** of future returns.

4. Leasing vs. Buying Decisions

Context:

A firm must decide whether to **lease** a machine or **purchase** it outright using borrowed funds.

- Leasing involves fixed monthly payments.
- Buying may involve a loan with APR.

Problem:

Leasing appears cheaper based on APR, but due to **monthly payments**, the **effective cost may be higher**.

Solution:

Calculate:

- **Present Value (PV)** of lease payments using **EAR** as discount rate.
- Compare to cost of buying (loan payments or outright purchase).

If PV of lease payments < buying cost, leasing is preferable.

This is especially useful in:

- Asset-heavy industries (manufacturing, logistics)
- Projects with short life spans or quickly depreciating assets

🎯 Summary Table: Application of EAR and APR

Use Case	What You Do	Why EAR is Important
Loan Comparison	Compare real cost of two bank loans	APR may hide compounding effects
Investment Comparison	Choose between deposits/investments	Compounding increases actual return
Cash Flow Discounting	Use correct rate in NPV/IRR	Ensures accurate financial analysis
Lease vs. Buy	Analyze total cost of lease vs. purchase	Helps determine better financing

4. Loan Amortization Schedules

1. What is a Loan?

A **loan** is money you borrow from a bank or someone else, which you **agree to pay back over time**, usually with **interest**.

For example:
If you borrow ₹1,00,000 from a bank, they'll ask you to pay it back in small parts every month — **plus a little extra as interest**.

2. What is Loan Amortization?

Amortization is just a fancy word for **"gradually paying off a loan over time"** by making regular payments (monthly or yearly).

Each payment has **two parts**:

- **Interest**: Money you pay to the bank for giving you the loan.
- **Principal**: The actual loan amount that you are paying back.

3. What is a Loan Amortization Schedule?

It's a **table** or list that shows:

Month	Payment	Interest	Principal	Remaining Loan
1	₹5,000	₹1,000	₹4,000	₹96,000
2	₹5,000	₹960	₹4,040	₹91,960
...	...	...	...	...

It tells you:

- How much you pay each month
- How much is going to interest
- How much is going to reduce the loan
- How much of the loan is still left

4. Why Does the Interest Reduce Over Time?

In the beginning, the **loan amount is high**, so the **interest is high**.

As you keep paying the loan:

- The **loan amount gets smaller**
- So, the **interest also becomes smaller**
- That means **more of your payment goes toward the principal**

5. Simple Example

Imagine you borrow ₹60,000 from a bank for 1 year at 10% interest.

You decide to pay it back in 12 monthly payments (EMI).

Every month, you pay around **₹5,275**

Month	Payment	Interest	Principal	Balance
1	₹5,275	₹500	₹4,775	₹55,225
2	₹5,275	₹460	₹4,815	₹50,410
3	₹5,275	₹420	₹4,855	₹45,555
...	...	...	...	...
12	₹5,275	₹45	₹5,230	₹0

You can see:

- Each month, the interest gets smaller
- More of your money goes to pay the loan back

6. Why Is It Important in Corporate Finance?

Companies also take loans — but **for bigger things** like:

- Buying machines
- Building offices
- Investing in new projects

They need to **plan how to repay the loan** without affecting their business. An amortization schedule helps them:

7. Tools to Create Amortization Schedules

Even beginners can make it using:

- 📊 **Excel**: Use `PMT()` to calculate EMI
- 🧮 **Online Calculators**
- 💻 **Python or software** (for automation)